

Alec HOYLAND

PERSONAL DATA

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TECHNICAL DETAILS

Programming Languages	Bash/Zsh, Python, MATLAB®, Julia, $\LaTeX$
Deployment & Cloud	Docker, Kubernetes, Helm, AWS, HPCC (slurm, SGE), GitOps
Machine Learning	NLP, computer vision, Transformers, autoencoders, LLMs
MLOps	TensorBoard, Guild.ai, dask, ray, vLLM, FastAPI/OpenAI, Triton
Math & Science	Numerical simulation, signal processing, Bayesian statistics, computational neuroscience, biochemistry/biophysics
Security	US citizen, TOP SECRET clearance
Education	MS Neuroscience, BS Neuroscience, Biological Physics, Brandeis University

WORK EXPERIENCE

Apr 2025—Oct 2025 <i>Institution</i>	SENIOR DATA SCIENTIST AE Studio , Los Angeles, CA (Remote) End-to-end developed a novel AI-powered language learning app, scaling to tens of thousands of users with a 90% approval rating. Research into AI alignment via gradient routing. Generative AI tools and techniques for personalized learning experiences.
Nov 2023—Apr 2025 <i>Institution</i>	SENIOR APPLIED MACHINE LEARNING ENGINEER LEGION (formerly Yurts Technologies Inc.) , San Francisco, CA (Remote) Development and deployment of an AI-powered application suite using LLMs and associated technologies such as retrieval augmented generation (RAG). Deployed into classified and unclassified customer environments for TCV over \$17M. Performed novel research on LLM&RAG metrics. Won “Best Communications Solution” at BRAVO 11 classified hackathon. Fine-tuned and quantized LLMs using bootstrapped datasets for overall 10% improvement in generation quality and 30% improvement in inference metrics.
Oct 2021—Oct 2023 <i>Institution</i>	SENIOR RESEARCH SCIENTIST Clarifai Inc. , Wilmington, DE (Remote) R&D for public sector AI/ML. Developed and optimized object detection, object tracking, and dataset enrichment algorithms using PyTorch, Docker, Kubernetes. Developed and deployed solutions on bare-metal and cloud compute. Technical lead on a multimillion dollar ARR line-of-effort as part of Project MAVEN for USDI/NGA. Competed against and outperformed trillion-dollar companies in head-to-head competition. Implemented unsupervised data preprocessing that improved state-of-the-art by up to 35% average precision on some classes. Presented at collaboration events and industry conferences. Onboarded, assigned and monitored work of senior and junior research staff. Maintained documentation on best practices/workflow for development.
Sep 2020—Sep 2021 <i>Institution</i>	STAFF RESEARCH SCIENTIST Boston Fusion Corp. , Lexington, MA (Remote) R&D for public sector AI/ML. Developed rule-based AI and machine learning models for denoising radar, maritime surveillance, sensor fusion, and semantic classification. Successfully navigated SBIR projects to Phase III as technical lead. Primary author on SBIR/STTR proposals and scrum master.

Jul 2018—Aug 2020 <i>Principal Investigator</i> <i>Institution</i>	SCIENTIFIC PROGRAMMER & DATA ANALYST Michael Hasselmo, DPHIL Center for Systems Neuroscience , Boston University, Boston, MA Designed and maintained software tools to study neuromodulation in the entorhinal cortex of rodents, including automating data analysis on high-performance computing hardware, designing statistical and numerical models, integrating state-of-the-art machine learning for motion tracking, and automating data collection in experiments using Arduino microcontrollers. Published three papers, two conference papers, and developed a tech stack still used by the lab today. Built and administrated the website. Enrolled in graduate courses concurrently through tuition remission.
Feb 2015—Jul 2018 <i>Principal Investigator</i> <i>Institution</i>	RESEARCH ASSISTANT IN COMPUTATIONAL NEUROSCIENCE Eve Marder, PhD Brandeis University, Waltham, MA Built software for neuronal simulation, including <code>xolotl</code> , which simulates at state-of-the-art speeds with built-in parameter optimization and real-time visualization. Software and documentation design patterns developed for <code>xolotl</code> have been adopted company-wide at Inscopix (VC-backed neurotech firm, \$20-25M ARR, acquired by Bruker Corp. in 2022). Studied neuromodulation using biophysically-realistic models of a motor circuit.

EDUCATION

Jan 2021-Feb 2024 <i>Institution</i> <i>Advisor</i> <i>Research</i>	PHD STUDENT IN BIOMEDICAL ENGINEERING Worcester Polytechnic Institute , Worcester, MA GPA: 4.0/4.0 Prof. Adam Lammert, PhD Using compressive sensing and machine learning to dramatically reduce the number of trials required to characterize high-dimensional representations of tinnitus in a reverse correlation experiment. Published in IEEE journals and at leading industry conferences. Coursework in neuroengineering, machine learning, and numerical linear algebra. As of Feb 2024, am on an indefinite leave of absence after my advisor left the university.
Aug 2018-Aug 2020 <i>Institution</i> <i>Courses Taken</i>	NON-DEGREE GRADUATE STUDENT IN ENGINEERING Boston University , Boston, MA GPA: 3.8/4.0 Linear Algebra, Statistical & Numerical Methods, Ordinary Differential Equations, Partial Differential Equations, Machine Learning, Universal Natural Language Processing, Artificial Intelligence.
May 2018 <i>Institution</i> <i>Thesis</i> <i>Advisor</i>	MASTER OF SCIENCE IN NEUROSCIENCE Brandeis University , Waltham MA Highest Honors, GPA: 3.4/4.0 <i>Differential Responses to Neuromodulation in Model Neurons of the Crustacean Stomatogastric Ganglion</i> Prof. Eve Marder, PhD
May 2018 <i>Institution</i> <i>Advisors</i>	BACHELOR OF SCIENCE IN NEUROSCIENCE BACHELOR OF SCIENCE IN BIOLOGICAL PHYSICS Brandeis University , Waltham MA Highest Honors, GPA: 3.4/4.0 Minor in the History of Ideas Prof. Eve Marder, PhD & Prof. Aparna Baskaran, PhD

PUBLICATIONS

- Hoyland, Alec, Nelson V. Barnett, Benjamin W. Roop, Danae Alexandrou, Myah Caplan, Jacob Mills, Benjamin Parrell, Divya A. Chari, and Adam C. Lammert (2023a). “Reverse Correlation Uncovers More Complete Tinnitus Spectra”. In: *IEEE Open Journal of Engineering in Medicine and Biology*. Conference Name: IEEE Open Journal of Engineering in Medicine and Biology, pp. 1–3. ISSN: 2644-1276. DOI: [10.1109/OJEMB.2023.3275051](https://doi.org/10.1109/OJEMB.2023.3275051).
- (Jan. 6, 2023b). *Reverse Correlation Uncovers More Complete Tinnitus Spectra*. Pages: 2022.12.23.521795 Section: New Results. DOI: [10.1101/2022.12.23.521795](https://doi.org/10.1101/2022.12.23.521795). URL: <https://www.biorxiv.org/content/10.1101/2022.12.23.521795v2> (visited on 01/10/2023).
- Dannenberg, Holger, Hallie Lazaro, Pranav Nambiar, Alec Hoyland, and Michael E Hasselmo (Dec. 10, 2020). “Effects of visual inputs on neural dynamics for coding of location and running speed in medial entorhinal cortex”. In: *eLife* 9. Ed. by Adrien Peyrache, Laura L Colgin, and Kevin Allen. Publisher: eLife Sciences Publications, Ltd, e62500. ISSN: 2050-084X. DOI: [10.7554/eLife.62500](https://doi.org/10.7554/eLife.62500). URL: <https://doi.org/10.7554/eLife.62500> (visited on 06/06/2022).
- Hasselmo, Michael E., Andrew S. Alexander, Alec Hoyland, Jennifer C. Robinson, Marianne J. Bezaire, G. William Chapman, Ausra Saudargiene, Lucas C. Carstensen, and Holger Dannenberg (Apr. 8, 2020). “The Unexplored Territory of Neural Models: Potential Guides for Exploring the Function of Metabotropic Neuromodulation”. In: *Neuroscience*. tex.ids: hasselmoUnexploredTerritoryNeuralInReview. ISSN: 0306-4522. DOI: [10.1016/j.neuroscience.2020.03.048](https://doi.org/10.1016/j.neuroscience.2020.03.048). URL: <http://www.sciencedirect.com/science/article/pii/S0306452220302141> (visited on 06/25/2020).
- Dannenberg, Holger, Craig Kelley, Alec Hoyland, Caitlin K. Monaghan, and Michael E. Hasselmo (May 1, 2019). “The Firing Rate Speed Code of Entorhinal Speed Cells Differs across Behaviorally Relevant Time Scales and Does Not Depend on Medial Septum Inputs”. In: *Journal of Neuroscience* 39.18. Citation Key Alias: dannenbergFiringRateSpeed2019a, pp. 3434–3453. ISSN: 0270-6474, 1529-2401. DOI: [10.1523/JNEUROSCI.1450-18.2019](https://doi.org/10.1523/JNEUROSCI.1450-18.2019). URL: <http://www.jneurosci.org/content/39/18/3434> (visited on 06/12/2019).
- Gorur-Shandilya, Srinivas, Alec Hoyland, and Eve Marder (2018). “Xolotl: An Intuitive and Approachable Neuron and Network Simulator for Research and Teaching”. In: *Frontiers in Neuroinformatics* 12. Publisher: Frontiers. ISSN: 1662-5196. DOI: [10.3389/fninf.2018.00087](https://doi.org/10.3389/fninf.2018.00087). URL: <https://www.frontiersin.org/articles/10.3389/fninf.2018.00087/full> (visited on 06/01/2021).
- Hoyland, Alec (2018). “Differential Responses to Neuromodulation in Model Neurons of the Crustacean Stomatogastric Ganglion”. Citation Key Alias: hoylandDifferentialResponsesNeuromodulation2018a. Thesis. Brandeis University. URL: <http://bir.brandeis.edu/handle/10192/35686> (visited on 08/14/2019).

CONFERENCE PAPERS & POSTERS

- Hoyland, Alec, Nelson Barnett, Benjamin Roop, Danae Alexandrou, Benjamin Parrell, Divya Chari, and Adam Lammert (Feb. 11, 2023). “Characterizing Complex Tinnitus Sounds Using Reverse Correlation: A Feasibility Study”. In: *Association for Research in Otolaryngology*. 46th Annual Midwinter Meeting. Orlando FL.
- Ning, Wing, John H. Bladon, Jerry Chen, S Steinwenter, Alec Hoyland, and Michael E. Hasselmo (2019). “A cortical-hippocampal network supporting the temporal organization of memory”. In: *2019 Neuroscience Meeting Planner*. Society for Neuroscience. 164.05. Chicago, IL.
- Dannenberg, Holger, Craig Kelley, Alec Hoyland, Caitlin K. Monaghan, and Michael E. Hasselmo (2018). “Speed coding by entorhinal cortex speed cells differs across behaviorally relevant timescales and is independent of cholinergic modulation”. In: *Society for Neuroscience*. 508.27. San Diego, CA.

MAINTAINED OPEN-SOURCE PROJECTS

Feb 2022-Present <i>Website</i>	VARIOUS ML DEMOS Diffusion Model Demo , Audio ML Demo , Multilayer Perceptron Demo Interactive demos of machine learning models, built using <code>streamlit</code> and <code>pytorch</code> , including generative image modeling using diffusion models, audio processing, denoising, and transcription using WHISPER, and a feedforward neural network demo for a middle-school robotics club.
Dec 2022-Present <i>Website</i> <i>Publications</i>	TINNITUS RECONSTRUCTION https://alec-hoyland.github.io/tinnitus-reconstruction/ Hoyland, N. Barnett, et al. 2023 Reconstructs a frequency spectrum representation of the subjective internal experience of tinnitus, using human subject data from a brief alternate-forced choice task. The reconstruction algorithms are ML- and compressed sensing-based.
Dec 2019-Present <i>Website</i>	NEURAL DECODER https://github.com/hasselmonians/neural-decoder Produces a stochastic convolutional model of the relationship between an extrinsic observative signal and an intrinsic covarying spike train.
Aug 2018-Present <i>Publications</i> <i>Website</i>	BANDWIDTHESTIMATOR Dannenberg, Kelley, et al. 2019; Dannenberg, Lazaro, et al. 2020 https://github.com/hasselmonians/BandwidthEstimator Implements a maximum-likelihood leave-one-out cross-validated bandwidth parameter estimation algorithms for general point processes (including spike trains).
Aug 2018-Present <i>Publications</i> <i>Website</i>	RATCATCHER Dannenberg, Kelley, et al. 2019; Dannenberg, Lazaro, et al. 2020 https://github.com/hasselmonians/RatCatcher A data- & analysis-agnostic pipeline for automating analysis on a high-performance computing cluster, with a local interface. Written in MATLAB®.
Aug 2017-Present <i>Publications</i> <i>Website</i>	XOLOTL Gorur-Shandilya, Hoyland, and Marder 2018; Hasselmo et al. 2020 https://go.brandeis.edu/xolotl A fast and flexible neuronal simulator in C++ with an extensive MATLAB® front-end. Achieves state-of-the-art efficiency on non-branching models.

FUNDING

Jan 2022	Tinnitus Characterization using Reverse Correlation with Applications to Retraining Therapies, Pilot Project Program, UMASS Center for Clinical and Translational Science (\$50,000)
Mar 2017	Computational Neuroscience Traineeship, NIH/NIMH (\$11,000)
Apr 2016	Quantitative Biology Research Community Fellowship, HHMI (\$5,000)